Title: **Evaluating the potential of applying remote sensing technology in Micro Plastics detection in the atmosphere.**

Movement of plastic wastes has no boundaries and hence plastic pollution is considered to a major threat to our planet[1]. It is also easily transported by wind and water due to its low density, complicating plastic waste management. Plastics disposal mismanagement causes their spread in the environments where they are brittle and fragmented (by the effect of heat, UV rays, winds, waves), forming microplastics (MPs), defined as plastic particles less than 5 mm in diameter. MPs are ubiquitous in a broad type of environments from marine, freshwater, to the atmosphere [2]. Although marine MPs' research remains at the forefront atmosphere continues underrepresented, so more efforts are still needed in this area. The need of standardized methods in sampling and measuring MPs in the atmosphere ensuring the reproducibility and comparability of results has been stressed in some studies [3]. Hence, finding new methodologies in assessing the type, quantity, and the behavior of MPs in the atmosphere is a matter of great concern. Emerging technologies such as remote sensing is currently being evaluated as a first order means for the detection, to its efficiency to cover a large area without human operations, its cost-effectiveness and the capability to monitor areas over time[4]. Hence it is beneficial that evaluating the potential of applying remote sensing technology in MPs detection in the atmosphere.

Plastics are hydrocarbon composites materials that share common absorption wavelength at 1730 and 2310 nanometers independently of the plastic type. As such, it is possible to use hydrocarbon detection indices to detect plastic components [5]. Furthermore, there are a few attempts made in using hyperspectral imagery for detecting varieties of polymers. Some attempts on plastic debris detection have determined that it might be possible to use the MODIS Aqua satellite to detect ocean plastic debris floating on the surface based on their reflectance values.[4]

Furthermore, spectral unmixing has proved to be a powerful method for plastic detection. However, the main driving element for the success of this method is the choice of endmembers. Plastic debris, whether they are floating or on the ground, are usually smaller than the pixel size and are therefore not large enough to form pure pixels. However, the interpolation can be used to estimate a pure plastic pixel even though it induces uncertainties regarding plastic presence at subpixel scale [5].

Remote sensing data has proved its potential and usability in monitoring environmental issues. In the past decade, remote sensing data has been used to identify, quantify, and detect plastic litter in the terrestrial and marine environment [6]. For example, remote sensing data is used to quantify agricultural plastic waste, terrestrial aggregated plastic waste, and marine plastic litter “[6]”. Novelli and Tarantino (2015) used the spectral band to separate the plastic litter from vegetation, sand, and water in the terrestrial environment using the Landsat 8 OLI/TIRS spectral bands[7]. Goncalves et al. (2020) used high-resolution aerial photographs for mapping plastic litter on Cabedelo beach dune systems” [6]”.

Based on the above background there is a potential of applying remote sensing technology in plastic detection in the environment. And it has already been applied in some scenarios such as marine debris monitoring. In this sense, the unique absorption values and reflectance values of plastics and the possibility to use spectral unmixing algorithms in GIS technology are positive directions towards the applicability. Therefore, it would be viable experimenting new remote technique methodologies to detect micro plastics in the atmosphere.

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